

ALI ELHELISY

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SUMMARY

Computer Engineering student with practical experience building full-stack web applications, relational database systems, object-oriented software, and low-level systems programs. Proficient across multiple languages and frameworks including Python, Java, C++, C#, and JavaScript. Experienced in REST API design, MVC architecture, SQL Server database design, and software engineering practices including testing, documentation, and version control.

EDUCATION

Bachelor of Science in Computer Engineering

Expected: [Year]

[University Name]

TECHNICAL SKILLS

Languages:	Python, C++, C, Java, C#, JavaScript, T-SQL
Web Frameworks:	Flask, ASP.NET Core MVC, Node.js, Express, Jinja2, HTML, CSS
Databases:	Microsoft SQL Server, relational schema design, EER/UML modeling, triggers
Concepts & Tools:	Object-oriented programming, REST API design, MVC, CRUD, Swagger, Git, GitHub, pytest, software analysis and design, multithreading, parallel computing, computer networking

PROJECTS

Football Scouting Web Application — Software Engineering

Python, Flask, Microsoft SQL Server, Jinja2, HTML, CSS, pytest

- Built a multi-user web application for managing football players, scouting reports, user accounts, and admin controls using Flask and SQL Server.
- Implemented role-based authentication (admin and scout roles) with session management and access control across all routes.
- Developed full CRUD functionality for players and scouting reports, plus server-side search and filtering.
- Wrote an automated test suite using pytest to verify application logic and API endpoints.

Football Scouting Web Application — Web Programming

C#, ASP.NET Core MVC, Microsoft SQL Server, HTML, CSS

- Re-implemented the scouting platform using ASP.NET Core MVC, applying the Model-View-Controller pattern with Razor views and model validation.

- Built features for player management, scouting reports, position filters, user accounts, and admin/scout role separation.

Football Scouting Web Application — System Analysis and Design

Node.js, Express, Microsoft SQL Server, Swagger, JavaScript

- Designed and implemented a REST API backend with Node.js and Express, fully documented using Swagger/OpenAPI specification.
- Produced system analysis and design deliverables including use-case diagrams, ER models, and workflow documentation.

Football Scouting Application — Object-Oriented Programming

Java, Swing, OOP

- Developed a desktop GUI application using Java Swing, applying OOP principles — encapsulation, inheritance, and polymorphism — to model players, scouts, reviews, and roles.
- Implemented local data persistence and role-based access within a clean class hierarchy.

Premier League Database Management System

T-SQL, Microsoft SQL Server

- Designed a normalized relational database schema covering clubs, players, matches, seasons, and statistics, including primary/foreign key constraints and referential integrity rules.
- Implemented database triggers, complex analytical queries, and set operations; produced full UML/EER documentation.

DNS/HTTP Network Communication Simulation

C++, Winsock API, Networking

- Simulated DNS name resolution over UDP and HTTP communication over TCP using the Windows Winsock API.
- Implemented shortest-path routing algorithms and measured QoS metrics including latency, BER, and packet-loss behavior.

Multithreading Performance Analysis

C++, `std::thread`, Parallel Computing

- Benchmarked single-threaded versus multi-threaded matrix multiplication using `std::thread` and the chrono library for high-resolution timing.
- Analyzed thread speedup, overhead costs, and optimal thread-count trade-offs under different workload sizes.